

# Matthew Butrovich

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## Summary

Database management systems (DBMS) expert with a deep understanding of high-performance distributed data processing. Strong background in transactional and analytical DBMS design (query execution, storage management, concurrency control, data formats), data lakes (Iceberg, Parquet), Linux internals (eBPF, networking), storage systems (distributed, devices, file systems), performance engineering (benchmarking, observability), systems for machine learning, and machine learning for systems. Committed to lifelong learning, peer mentorship, and contributing to scientific and open source communities.

## Professional Experience

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|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2024 –      | <b>Apple</b><br>Senior Distributed Database Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Washington, DC |
|             | <ul style="list-style-type: none"><li>Enhancing Apple’s data lakehouse by contributing to multiple open source Apache projects: Arrow, DataFusion, DataFusion Comet, Iceberg, Parquet, and Spark.</li><li>Brought Comet to production quality, reducing the cost of Spark pipelines by &gt;50%.</li><li>Enabled Comet’s dynamic partition pruning (adaptive and non-adaptive query execution), which previously forced queries to fall back to slower Spark execution. Across 78 TPC-DS benchmark queries, the share of each query running natively rose from ~30–50% to ~95–97%, contributing to a &gt;50% overall speedup at 1 TB.</li><li>Designed and shipped Comet’s native Iceberg scan, upstreaming &gt;30 PRs to Iceberg.</li><li>Drove a DataFusion sort-merge-join (SMJ) overhaul across ~12 PRs, making TPC-H &gt;500x faster and near-unique LEFT/FULL joins &gt;20x faster.</li><li>Diagnosed and fixed a Comet busy-poll regression, dropping cluster CPU utilization from 30% to 8% on I/O-bound Iceberg workloads.</li><li>Re-architected the JVM-to-native data path to true zero-copy using the Arrow C Stream Interface, eliminating a per-batch deep copy on every batch crossing the boundary.</li></ul> |                |
| 2009 – 2017 | <b>Western Digital Corporation</b><br>Lead Technician V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Irvine, CA     |
|             | <ul style="list-style-type: none"><li>Led a team of 17 technicians verifying firmware for HDD and SSD storage devices.</li><li>Promoted four times from initial role as Technician II while earning my B.Sc.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |

## Education

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|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2019 – 2024 | <b>Carnegie Mellon University</b><br>Doctor of Philosophy (Ph.D.), Computer Science                                                                                                                              | Pittsburgh, PA |
|             | <ul style="list-style-type: none"><li>Advisor: <a href="#">Andy Pavlo</a></li><li>Thesis: “On Embedding Database Management System Logic in Operating Systems via Restricted Programming Environments”</li></ul> |                |
| 2017 – 2018 | <b>Carnegie Mellon University</b><br>Master of Science (M.Sc.), Computer Science                                                                                                                                 | Pittsburgh, PA |
| 2013 – 2017 | <b>University of California, Irvine</b><br>Bachelor of Science (B.Sc.), <i>cum laude</i> , Computer Science and Engineering                                                                                      | Irvine, CA     |
| 2010 – 2013 | <b>Irvine Valley College</b><br>Certificate of Achievement, Computer Science: Computer Languages                                                                                                                 | Irvine, CA     |

## Selected Talks

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|------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 2025 | <b>CS 224P: Big Data Systems</b><br>“Fusing Academic Research and Open Source Systems for Petabyte-Scale Analytics” | <a href="#">Link</a> <a href="#">YouTube</a> |
| 2025 | <b>DC Systems</b><br>“Putting Theory Into Practice”                                                                 | <a href="#">Link</a> <a href="#">YouTube</a> |
| 2023 | <b>15-445/645: Database Systems</b><br>Lecture on join algorithms                                                   | <a href="#">Link</a> <a href="#">YouTube</a> |
| 2019 | <b>15-445/645: Database Systems</b><br>Lecture on two-phase locking concurrency control                             | <a href="#">Link</a> <a href="#">YouTube</a> |

## Selected Publications

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| 2025 | <b>Matthew Butrovich</b> , Samuel Arch, Wan Shen Lim, William Zhang, Jignesh M. Patel, and Andrew Pavlo. “BPF-DB: A Kernel-Embedded Transactional Database Management System For eBPF Applications”. <i>SIGMOD Conference</i> .                                         |
| 2023 | <b>Matthew Butrovich</b> , Karthik Ramanathan, John Rollinson, Wan Shen Lim, William Zhang, Justine Sherry, and Andrew Pavlo. “Tigger: A Database Proxy That Bounces With User-Bypass”. <i>Proc. VLDB Endow.</i>                                                        |
| 2022 | <b>Matthew Butrovich</b> , Wan Shen Lim, Lin Ma, John Rollinson, William Zhang, Yu Xia, and Andrew Pavlo. “Tastes Great! Less Filling! High Performance and Accurate Training Data Collection for Self-Driving Database Management Systems”. <i>SIGMOD Conference</i> . |

## Service

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### To the Open Source Community

|             |                                                                         |                      |
|-------------|-------------------------------------------------------------------------|----------------------|
| 2026 –      | PMC Member, Apache DataFusion, <i>The Apache Software Foundation</i> .  | <a href="#">Link</a> |
| 2025 –      | Contributor, Apache Arrow, <i>The Apache Software Foundation</i> .      | <a href="#">Link</a> |
| 2025 –      | Contributor, Apache Iceberg, <i>The Apache Software Foundation</i> .    | <a href="#">Link</a> |
| 2025 –      | Contributor, Apache Parquet, <i>The Apache Software Foundation</i> .    | <a href="#">Link</a> |
| 2024 –      | Contributor, Apache Spark, <i>The Apache Software Foundation</i> .      | <a href="#">Link</a> |
| 2025 – 2026 | Committer, Apache DataFusion, <i>The Apache Software Foundation</i> .   | <a href="#">Link</a> |
| 2024 – 2025 | Contributor, Apache DataFusion, <i>The Apache Software Foundation</i> . | <a href="#">Link</a> |

### To the Profession

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|------|-------------------------------------------------------------------------------|----------------------|
| 2026 | Member, <i>ACM SIGMOD 2027 Program Committee</i> .                            | <a href="#">Link</a> |
| 2026 | Member, <i>VLDB 2027 Program Committee</i> .                                  | <a href="#">Link</a> |
| 2026 | Member, <i>4th Workshop on eBPF and Kernel Extensions Program Committee</i> . | <a href="#">Link</a> |
| 2025 | Member, <i>ACM SIGMOD 2026 Industry Track Program Committee</i> .             | <a href="#">Link</a> |
| 2025 | Member, <i>ACM SIGMOD 2026 Program Committee</i> .                            | <a href="#">Link</a> |
| 2025 | Reviewer, <i>ACM Transactions on Database Systems (TODS)</i> .                | <a href="#">Link</a> |
| 2024 | Member, <i>eBPF Summit 2024 Program Committee</i> .                           | <a href="#">Link</a> |
| 2023 | Member, <i>eBPF Summit 2023 Program Committee</i> .                           | <a href="#">Link</a> |
| 2023 | Member, <i>ACM SIGMOD 2023 Artifacts &amp; Reproducibility Committee</i> .    | <a href="#">Link</a> |

## Skills

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**Programming Languages:** C, C++, eBPF, Java, Python, Rust, Scala

**Development Tools:** bpftrace, GDB/LLDB, Git, Intel VTune, perf